Problem 1. Consider the function $d\colon \mathbb{N} \to \mathbb{N}$ defined $d(n) =$ the number of divisors of n. For example, $d(12) = 6$ (because the divisors of 12 are $1, 2, 3, 4, 6, 12$).

1. Let A be the set of all natural numbers n such that $d(n) = 2$. (I.e., A is the *preimage* of 2.) Determine what A is.
2. Prove or disprove the following statement.

   Let $m, n \in \mathbb{N}$. If $m \neq n$, then $d(m) \neq d(n)$.

3. Let $n \in \mathbb{N}$. Prove that if a natural number a divides 2^{n+1}, then either $a = 2^{n+1}$ or a divides 2^n.
4. Find a formula for $d(2^n)$ for all $n \in \mathbb{N}$ and prove that your formula is correct.
5. Is d a surjective function? (Prove your answer.)

Problem 2. Let $M_2 = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \;\middle|\; a, b, c, d \in \mathbb{R} \right\}$, i.e. M_2 is the set of 2×2 matrices with entries in $\mathbb{R}$. We define a function $T\colon M_2 \to M_2$ by

$$T\left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \right\} = \begin{bmatrix} a & c \\ b & d \end{bmatrix}.$$

We call T the **transpose** function. Answer the following, and justify your answers completely.

1. Is T injective?
2. Is T surjective?
3. Is T bijective?
4. Let $S \subseteq M_2$ be the set such that $T(X) = X$ for all $X \in S$. Describe S as explicitly as possible.
5. Let $U = \left\{ \begin{bmatrix} a & b \\ 0 & d \end{bmatrix} \;\middle|\; a, b, d \in \mathbb{R} \right\}$. Describe $T^{-1}(U)$ as explicitly as possible.

Problem 3. Assume A and B are nonempty sets and let $f\colon A \to B$. Prove or disprove the following.

1. For all $U \subseteq A$ and $V \subseteq A$, $f(U \cup V) = f(U) \cup f(V)$.
2. For all $X \subseteq B$ and $Y \subseteq B$, $f^{-1}(X \cap Y) = f^{-1}(X) \cap f^{-1}(Y)$.

Problem 4. Let A, B, and C be nonempty sets and $f\colon A \to B$.

1. Assume that f is surjective. Show that if $g\colon B \to C$ and $h\colon B \to C$ are functions such that $g \circ f = h \circ f$, then $g = h$.
2. Show that Part 1 is not necessarily true if f is not surjective.

Problem 5. Let $f\colon A \to B$ and $g\colon B \to C$. Prove that if $g \circ f$ is a surjection, then g is a surjection. Give a counterexample to show that the converse does not hold.

Problem 6. Let $f\colon A \to B$ and $g\colon B \to C$. Prove that if f and g are both injective, then $g \circ f$ is injective. Give a counterexample to show that the converse does not hold.

Problem 7. Do not turn these in, but complete the progress checks in the sections of textbook.